K-mer MLP for Full GO Prediction

Project goal

We train a multi-label MLP model to predict Gene Ontology (GO) terms from k-mer frequency features extracted from protein sequences.

This experiment uses the **full GO label space**.
The goal is to build a complete baseline that reflects the real difficulty of CAFA-style protein function prediction:

- high-dimensional output space
- severe class imbalance
- extremely sparse labels
- multi-label prediction for each protein

Why this notebook matters

Our data preparation pipeline already produced:

- cleaned and aligned protein samples
- normalized k-mer feature vectors
- train/validation splits
- multi-label GO targets stored as sparse matrices

This begins at the **model construction stage**, using the processed k-mer outputs as direct inputs for supervised learning.

Input / output summary

- **Input features:** normalized 2-mer frequency vectors, dimension = 400
- **Output labels:** all GO terms, dimension = 31,454
- **Task type:** multi-label classification
- **Evaluation:** validation loss and threshold-based F1 analysis

Step 0. Imports

In [ ]:
```
import os
import json
import pickle
import numpy as np
import matplotlib.pyplot as plt
```

```
from scipy.sparse import load_npz

import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import Dataset, DataLoader

from sklearn.metrics import f1_score
```

Step 1. Load Processed K-mer Training Data

Why this step?

The k-mer preprocessing notebook already converted raw protein sequences into standardized 2-mer frequency vectors and saved the aligned GO labels.

Loading these processed files allows us to begin model training directly, without repeating:

- sequence cleaning
- GO label alignment
- k-mer vectorization
- train/validation splitting
- feature standardization

Files loaded

We load:

- `X_train.npy` and `X_val.npy` : standardized k-mer feature matrices
- `Y_train.npz` and `Y_val.npz` : sparse multi-label GO target matrices
- `train_idx.npy` and `val_idx.npy` : split indices
- `meta.pkl` : metadata including vocabulary, scaler, protein IDs, and GO encoder

Why sparse labels matter

The target matrix contains all GO terms, so the label space is very large and mostly zeros.
Keeping the targets in sparse form helps avoid unnecessary memory use.

In [ ]:
```
DATA_DIR = "data_processed_kmer"

X_train = np.load(os.path.join(DATA_DIR, "X_train.npy"))
X_val   = np.load(os.path.join(DATA_DIR, "X_val.npy"))

Y_train = load_npz(os.path.join(DATA_DIR, "Y_train.npz"))
Y_val   = load_npz(os.path.join(DATA_DIR, "Y_val.npz"))
```

```
train_idx = np.load(os.path.join(DATA_DIR, "train_idx.npy"))
val_idx   = np.load(os.path.join(DATA_DIR, "val_idx.npy"))

with open(os.path.join(DATA_DIR, "meta.pkl"), "rb") as f:
    meta = pickle.load(f)

print("X_train:", X_train.shape, X_train.dtype)
print("X_val:  ", X_val.shape, X_val.dtype)
print("Y_train:", Y_train.shape, type(Y_train))
print("Y_val:  ", Y_val.shape, type(Y_val))
print("train_idx:", train_idx.shape)
print("val_idx:  ", val_idx.shape)
print("meta keys:", list(meta.keys()))
```

Step 2. Sanity Check of Input and Output Dimensions

Why this step?

Before defining the model, we verify that:

1. the input dimension matches the expected k-mer representation
2. the output dimension matches the full GO label space
3. the metadata is aligned with the loaded arrays

This step helps catch common issues early, such as:

- mismatched feature files
- wrong metadata version
- incorrect GO encoder loading
- accidental use of filtered labels instead of full labels

Expected dimensions

From the data preparation notebook, the k-mer feature dimension should be **400**, and the full GO output dimension should be **31,454**. This matches the CAFA5 preprocessing summary.

In [ ]:
```
print("Input dim (k-mer):", X_train.shape[1])
print("Output dim (all GO terms):", Y_train.shape[1])

if "mlb" in meta:
    mlb = meta["mlb"]
    print("Number of GO classes from mlb:", len(mlb.classes_))
    print("First 10 GO terms:", mlb.classes_[:10])

if "protein_ids" in meta:
    protein_ids = meta["protein_ids"]
    print("Number of protein_ids:", len(protein_ids))
```

Step 3. Build a Sparse-Safe PyTorch Dataset

Why this step?

PyTorch models expect mini-batches of tensors, but our GO label matrix is stored as a sparse matrix to save memory.

If we convert the full target matrix into a dense array all at once, memory usage can become unnecessarily large because:

- the number of proteins is large
- the GO label space contains over 31,000 classes
- the label matrix is extremely sparse

Design choice

We therefore define a custom dataset that:

- reads dense k-mer features from NumPy arrays
- converts only one sparse label row at a time into a dense tensor
- returns `(x, y)` pairs ready for mini-batch training

Why this is important

This lets us train on the **full GO label space** without filtering labels and without densifying the entire target matrix in memory.

In [ ]:
```
class ProteinKmerDataset(Dataset):
    def __init__(self, X, Y_sparse):
        self.X = X
        self.Y = Y_sparse

    def __len__(self):
        return self.X.shape[0]

    def __getitem__(self, idx):
        x = torch.tensor(self.X[idx], dtype=torch.float32)
        y = torch.tensor(self.Y[idx].toarray().ravel(), dtype=torch.float32)
        return x, y
```

Step 4. Create DataLoaders for Mini-Batch Training

Why this step?

DataLoaders provide efficient mini-batch iteration during training and validation.

We use them to:

- shuffle training samples each epoch
- batch proteins into manageable groups
- keep validation order fixed
- optionally use pinned memory for faster GPU transfer

Why batch size is kept moderate

Even though the input dimension is small (400), the output dimension is very large (31,454).
This means each forward/backward pass still produces large output tensors, so a moderate batch size is safer for memory and runtime stability.

In [ ]:
```
train_dataset = ProteinKmerDataset(X_train, Y_train)
val_dataset   = ProteinKmerDataset(X_val, Y_val)

batch_size = 64

train_loader = DataLoader(
    train_dataset,
    batch_size=batch_size,
    shuffle=True,
    num_workers=0,
    pin_memory=torch.cuda.is_available()
)

val_loader = DataLoader(
    val_dataset,
    batch_size=batch_size,
    shuffle=False,
    num_workers=0,
    pin_memory=torch.cuda.is_available()
)

print("Train batches:", len(train_loader))
print("Val batches:  ", len(val_loader))
```

Step 5. Define the K-mer MLP Model

Why this step?

The model maps each protein's k-mer frequency vector to a multi-label GO prediction vector.

Because the input features are fixed-length tabular vectors, a multilayer perceptron (MLP) is a natural baseline:

- simple to implement
- efficient to train

- appropriate for dense handcrafted features like k-mer frequencies

Model structure

The architecture uses:

- two hidden layers
- batch normalization for optimization stability
- ReLU nonlinearities
- dropout for regularization

Why an MLP is a reasonable baseline

The k-mer representation captures local amino acid composition but not full sequence context.
An MLP provides a clean baseline for testing how much predictive signal exists in these handcrafted sequence features before moving to richer representations such as ESM embeddings or structural graphs. This is consistent with the role of k-mer features in your preprocessing notebook, where they are described as a fast and interpretable baseline.

In [ ]:
```
class KmerMLP(nn.Module):
    def __init__(self, input_dim, output_dim, hidden1=512, hidden2=256, drop
        super().__init__()
        self.net = nn.Sequential(
            nn.Linear(input_dim, hidden1),
            nn.BatchNorm1d(hidden1),
            nn.ReLU(),
            nn.Dropout(dropout),

            nn.Linear(hidden1, hidden2),
            nn.BatchNorm1d(hidden2),
            nn.ReLU(),
            nn.Dropout(dropout),

            nn.Linear(hidden2, output_dim)
        )

    def forward(self, x):
        return self.net(x)
```

Step 6. Initialize the Model on the Available Device

Here we:

- detect whether a GPU is available
- construct the model using the correct input and output dimensions
- move the model to the computation device

- inspect the total number of trainable parameters

Why parameter count matters

The input dimension is modest, but the output layer is large because the model predicts all GO terms at once.
Printing the parameter count helps us understand the memory and optimization cost of training on the full label space.

In [ ]:
```
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print("Device:", device)

model = KmerMLP(
    input_dim=X_train.shape[1],   # 400
    output_dim=Y_train.shape[1]   # 31454
).to(device)

n_params = sum(p.numel() for p in model.parameters())
print("Total parameters:", f"{n_params:,}")
print(model)
```

Step 7. Compute Class Weights for Extreme Label Imbalance

Why this step?

GO prediction is an extremely imbalanced multi-label problem.

In the preprocessing analysis, the label matrix was shown to be highly sparse, with most entries equal to zero and many GO terms appearing only rarely. The long-tail frequency pattern is one of the main challenges of this task.

If we train with an unweighted binary loss, the model can minimize loss by predicting negatives for most labels.
This would lead to poor learning for rare positive GO terms.

What we do

We compute a `pos_weight` value for each GO term:

- rare labels receive higher positive weight
- common labels receive lower positive weight

Why BCEWithLogitsLoss + pos_weight

This is the standard choice for multi-label classification when:

- each output is an independent binary target
- the label distribution is heavily imbalanced
- we want the model to pay more attention to rare positive examples

In [ ]:
```
label_counts = np.asarray(Y_train.sum(axis=0)).ravel().astype(np.float32)
n_train = Y_train.shape[0]

# Compute positive class weights for each GO term.
# Rare labels receive larger weights, while common labels receive smaller we
# Add a small epsilon to avoid division by zero for labels with no positive
pos_weight = (n_train - label_counts) / (label_counts + 1e-6)

# Clip extreme values to keep training numerically stable.
# Without clipping, very rare labels can produce huge weights and explode th
pos_weight = np.clip(pos_weight, 1.0, 100.0)

print("label_counts shape:", label_counts.shape)
print("pos_weight shape:", pos_weight.shape)
print("label count min/max:", label_counts.min(), label_counts.max())
print("pos_weight min/max:", pos_weight.min(), pos_weight.max())

# Convert to a PyTorch tensor on the same device as the model.
pos_weight_tensor = torch.tensor(pos_weight, dtype=torch.float32, device=dev

# Use weighted binary cross-entropy for multi-label classification.
criterion = nn.BCEWithLogitsLoss(pos_weight=pos_weight_tensor)

# Adam optimizer with light L2 regularization.
optimizer = optim.Adam(model.parameters(), lr=1e-3, weight_decay=1e-5)
```

Step 8. Define Training and Validation Loops

Why this step?

To train the model reproducibly, we separate the workflow into two functions:

- `train_one_epoch(...)` for parameter updates
- `evaluate_loss(...)` for validation without gradient computation

Why this separation helps

This makes the notebook easier to:

- debug
- reuse for ESM features later
- compare training vs validation behavior
- save the best model based on validation performance

What happens during training

For each batch:

1. move inputs and targets to the device
2. compute logits
3. compute the weighted multi-label loss
4. backpropagate gradients
5. update model parameters

In [ ]:
```
def train_one_epoch(model, loader, criterion, optimizer, device):
    model.train()
    total_loss = 0.0

    for X_batch, Y_batch in loader:
        X_batch = X_batch.to(device, non_blocking=True)
        Y_batch = Y_batch.to(device, non_blocking=True)

        optimizer.zero_grad()
        logits = model(X_batch)
        loss = criterion(logits, Y_batch)
        loss.backward()
        optimizer.step()

        total_loss += loss.item() * X_batch.size(0)

    return total_loss / len(loader.dataset)


@torch.no_grad()
def evaluate_loss(model, loader, criterion, device):
    model.eval()
    total_loss = 0.0

    for X_batch, Y_batch in loader:
        X_batch = X_batch.to(device, non_blocking=True)
        Y_batch = Y_batch.to(device, non_blocking=True)

        logits = model(X_batch)
        loss = criterion(logits, Y_batch)

        total_loss += loss.item() * X_batch.size(0)

    return total_loss / len(loader.dataset)
```

Step 9. Collect Validation Predictions

Why this step?

Loss alone does not fully describe model quality in a multi-label setting.
To evaluate the classifier meaningfully, we need predicted probabilities for every GO term on the validation set.

What this function returns

For all validation proteins, we collect:

- predicted probabilities after sigmoid
- true binary GO label vectors

These will later be used to:

- sweep over decision thresholds
- compute micro-F1 and macro-F1
- identify a threshold that gives the best prediction quality

Why probabilities are needed

In multi-label classification, a default threshold of 0.5 is often suboptimal, especially under heavy class imbalance.

In [ ]:
```
@torch.no_grad()
def get_predictions(model, loader, device):
    model.eval()
    all_probs = []
    all_targets = []

    for X_batch, Y_batch in loader:
        X_batch = X_batch.to(device, non_blocking=True)
        logits = model(X_batch)
        probs = torch.sigmoid(logits).cpu().numpy()

        all_probs.append(probs)
        all_targets.append(Y_batch.numpy())

    all_probs = np.vstack(all_probs)
    all_targets = np.vstack(all_targets)

    return all_probs, all_targets
```

Step 10. Evaluate Predictions Across Multiple Thresholds

Why this step?

Each GO term is predicted independently as a probability.
To convert probabilities into binary predictions, we must choose a threshold.

A fixed threshold such as 0.5 may not work well because:

- the label matrix is extremely sparse
- many true positives are low-confidence

- the optimal precision/recall tradeoff may occur at a lower threshold

Why threshold sweeping matters

The CAFA evaluation philosophy emphasizes threshold-based optimization of F-score. Your preprocessing report also notes that the official metric is weighted F-max, which is found by sweeping thresholds. :contentReference[oaicite:4]{index=4}

Metrics reported

We compute:

- **micro-F1**: emphasizes overall performance across all labels
- **macro-F1**: treats each label equally and reveals long-tail difficulty

In [ ]:
```
def threshold_sweep(y_true, y_prob, thresholds=None):
    if thresholds is None:
        thresholds = np.arange(0.05, 0.55, 0.05)

    records = []

    for t in thresholds:
        y_pred = (y_prob >= t).astype(np.int8)

        micro = f1_score(y_true, y_pred, average="micro", zero_division=0)
        macro = f1_score(y_true, y_pred, average="macro", zero_division=0)

        records.append({
            "threshold": float(t),
            "micro_f1": float(micro),
            "macro_f1": float(macro)
        })

        print(f"t={t:.2f} | micro-F1={micro:.4f} | macro-F1={macro:.4f}")

    best_micro_row = max(records, key=lambda x: x["micro_f1"])
    best_macro_row = max(records, key=lambda x: x["macro_f1"])

    return records, best_micro_row, best_macro_row
```

Step 11. Train the K-mer MLP Model

Why this step?

This is the main optimization stage of the notebook.

For each epoch, we:

1. train on the training set
2. compute validation loss

3. record both losses for visualization
4. save the model checkpoint with the best validation loss

Why save the best checkpoint

The best validation-loss model is usually a safer choice than the final epoch model, especially when overfitting may occur.

Why start with a small number of epochs

Because the full GO output space is large and expensive, it is practical to first confirm that:

- the pipeline runs correctly
- the loss decreases
- the model checkpointing works before scaling up to longer training.

In [ ]:
```
num_epochs = 3
save_path = "best_kmer_mlp_full_go.pt"

train_losses = []
val_losses = []

best_val_loss = float("inf")

for epoch in range(num_epochs):
    train_loss = train_one_epoch(model, train_loader, criterion, optimizer,
    val_loss = evaluate_loss(model, val_loader, criterion, device)

    train_losses.append(train_loss)
    val_losses.append(val_loss)

    print(f"Epoch {epoch+1:02d}/{num_epochs} | train_loss={train_loss:.6f} |

    if val_loss < best_val_loss:
        best_val_loss = val_loss
        torch.save(model.state_dict(), save_path)
        print(f"  -> saved best model to {save_path}")
```

Step 12. Visualize the Training Curve

Why this step?

A loss curve provides a quick diagnosis of model behavior.

By plotting training and validation loss across epochs, we can check for:

- successful optimization
- underfitting

- early signs of overfitting
- unstable training

What to look for

- training loss should generally decrease
- validation loss should ideally also decrease or stabilize
- a widening gap between them may suggest overfitting

In [ ]:
```
plt.figure(figsize=(6, 4))
plt.plot(range(1, num_epochs + 1), train_losses, marker='o', label="train")
plt.plot(range(1, num_epochs + 1), val_losses, marker='o', label="val")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.title("K-mer MLP training curve (full GO)")
plt.legend()
plt.show()
```

Step 13. Reload the Best Model and Generate Validation Outputs

Why this step?

Instead of evaluating the final epoch model, we reload the best checkpoint saved during training.
This ensures that the threshold sweep is performed on the model that achieved the best validation loss.

Why this matters

For fair model comparison later, we want evaluation to reflect the strongest saved version of the model, not just the latest one.

In [ ]:
```
best_model = KmerMLP(
    input_dim=X_train.shape[1],
    output_dim=Y_train.shape[1]
).to(device)

best_model.load_state_dict(torch.load(save_path, map_location=device))

val_probs, val_targets = get_predictions(best_model, val_loader, device)

print("val_probs shape:", val_probs.shape)
print("val_targets shape:", val_targets.shape)
```

Step 14. Run Threshold Sweep and Identify Best Validation Thresholds

Why this step?

This step translates validation probabilities into final binary predictions using multiple thresholds.

Rather than assuming one universal threshold in advance, we search over a range of values and record:

- the threshold that maximizes micro-F1
- the threshold that maximizes macro-F1

Why compare both

- **Best micro-F1** shows overall large-scale prediction quality
- **Best macro-F1** highlights how the model behaves on rare terms

In a long-tail problem like GO prediction, these two thresholds may not be the same.

In [ ]:
```
thresholds = np.arange(0.05, 0.55, 0.05)
records, best_micro_row, best_macro_row = threshold_sweep(val_targets, val_p

print("\nBest by micro-F1:", best_micro_row)
print("Best by macro-F1:", best_macro_row)
```

Step 15. Save Training Results and Evaluation Summary

Why this step?

Saving outputs to disk makes the experiment reproducible and easier to compare against future models such as:

- ESM-MLP
- namespace-specific models
- filtered-label variants
- structure-based graph models

What we save

We store:

- model name
- input and output dimensions
- batch size and epoch count
- best validation loss

- best threshold-based F1 results
- training and validation loss history

Why this is useful

This creates a lightweight experiment record that can be reused in reports, slides, or comparison tables later.

In [ ]:
```
results = {
    "model": "kmer_mlp_full_go",
    "input_dim": int(X_train.shape[1]),
    "output_dim": int(Y_train.shape[1]),
    "batch_size": batch_size,
    "num_epochs": num_epochs,
    "best_val_loss": float(best_val_loss),
    "best_micro": best_micro_row,
    "best_macro": best_macro_row,
    "train_losses": train_losses,
    "val_losses": val_losses,
}

with open("kmer_mlp_full_go_metrics.json", "w") as f:
    json.dump(results, f, indent=2)

print("Saved metrics to kmer_mlp_full_go_metrics.json")
```

In [ ]:
```
import pandas as pd

df_thresh = pd.DataFrame(records)
display(df_thresh)
```

Step 16. Visualize Micro-F1 and Macro-F1 Across Thresholds

Why this step?

A plot provides a more intuitive view of threshold sensitivity than a table alone.

What this figure helps us see

- where micro-F1 peaks
- where macro-F1 peaks
- whether the model is robust to small threshold changes
- whether optimizing for global performance differs from optimizing for rare-label balance

Interpretation note

For full-label GO prediction, it is common for macro-F1 to remain much lower than micro-F1 because many GO terms are extremely rare. This reflects the long-tail label distribution observed during preprocessing. :contentReference[oaicite:5]{index=5}

In [ ]:
```
plt.figure(figsize=(6, 4))
plt.plot(df_thresh["threshold"], df_thresh["micro_f1"], marker='o', label="m
plt.plot(df_thresh["threshold"], df_thresh["macro_f1"], marker='o', label="m
plt.xlabel("Threshold")
plt.ylabel("F1")
plt.title("Threshold sweep (full GO)")
plt.legend()
plt.show()
```

In [ ]:

ESM MLP for Full GO Prediction

Project goal

In this notebook, we train a multi-label MLP model to predict Gene Ontology (GO) terms from pretrained ESM protein embeddings.

This experiment uses the **full GO label space**, meaning we keep all GO terms rather than filtering to only frequent labels.
The purpose is to build a stronger sequence-based baseline than k-mer features while preserving the real difficulty of the CAFA-style prediction task:

- large output space
- severe class imbalance
- extremely sparse labels
- multi-label protein function prediction

Why this notebook matters

The earlier preprocessing workflow already:

- cleaned and aligned protein sequences
- generated ESM embeddings for each protein
- encoded GO terms into a multi-label target matrix
- saved metadata such as protein IDs and GO vocabularies

This notebook begins at the **supervised modeling stage**, using pretrained sequence representations as direct input to an MLP classifier.

Input / output summary

- **Input features:** mean-pooled ESM2 protein embeddings, dimension = 640
- **Output labels:** all GO terms, dimension = 31,454
- **Task type:** multi-label classification
- **Evaluation:** validation loss and threshold-based F1 analysis

Step 0. Imports and Modeling Setup

Why this step?

Before training the ESM-based model, we import all libraries needed for:

- loading ESM embeddings and GO labels
- handling sparse target matrices safely
- defining PyTorch datasets and neural networks
- optimizing and evaluating a multi-label classifier
- saving experiment outputs for later comparison

What we import

This notebook uses:

- **NumPy / SciPy** for arrays and sparse matrices
- **pickle / os / json** for metadata loading and result saving
- **PyTorch** for dataset loading, model definition, and training
- **scikit-learn** for feature standardization and F1 evaluation
- **Matplotlib** for training and threshold-sweep visualization

In [1]:
```
import os
import pickle
import numpy as np
from scipy.sparse import load_npz

DATA_DIR = "data_processed_esm"

X_esm = np.load(os.path.join(DATA_DIR, "X_esm.npy"))

print("X_esm shape:", X_esm.shape, X_esm.dtype)

# Try sparse Y first
y_sparse_path = os.path.join(DATA_DIR, "Y_sparse.npz")
y_dense_path = os.path.join(DATA_DIR, "Y.npy")

if os.path.exists(y_sparse_path):
    Y = load_npz(y_sparse_path)
    print("Loaded sparse Y:", Y.shape, type(Y))
elif os.path.exists(y_dense_path):
    Y = np.load(y_dense_path)
    print("Loaded dense Y:", Y.shape, type(Y))
else:
    raise FileNotFoundError("Cannot find Y_sparse.npz or Y.npy")

protein_ids_path = os.path.join(DATA_DIR, "protein_ids.pkl")
if os.path.exists(protein_ids_path):
    with open(protein_ids_path, "rb") as f:
        protein_ids = pickle.load(f)
    print("protein_ids length:", len(protein_ids))
```

```
X_esm shape: (140569, 640) float32
Loaded sparse Y: (140569, 31454) <class 'scipy.sparse._csr.csr_matrix'>
```

Step 1. Verify Data Alignment and Dimensionality

Why this step?

Before splitting the data or defining the model, we need to confirm that:

1. the number of ESM embedding rows matches the number of GO target rows
2. the embedding dimension is correct
3. the protein order has remained aligned throughout preprocessing

What we expect

From the preprocessing workflow, the ESM representation should produce one embedding per protein, with an embedding dimension of **320**. This matches the ESM2 setup described in the report. :contentReference[oaicite:1]{index=1}

Why alignment is critical

A mismatch between ESM features and GO labels would silently corrupt training.
In a supervised protein function prediction task, each row of the feature matrix must correspond to the exact same protein as the same row of the target matrix.

In [2]:
```
print("Number of ESM samples:", X_esm.shape[0])
print("Number of Y samples:", Y.shape[0])

assert X_esm.shape[0] == Y.shape[0], "X_esm and Y are not aligned!"
```

```
Number of ESM samples: 140569
Number of Y samples: 140569
```

Step 2. Reuse the Same Train/Validation Split as the K-mer Baseline

Why this step?

To compare ESM and k-mer models fairly, both models should be trained and evaluated on the exact same protein samples.

If each feature type uses a different random split, performance differences could reflect data differences rather than model quality.

What we do

We reuse the previously saved `train_idx` and `val_idx` arrays from the k-mer pipeline and apply them to the ESM embedding matrix.

Why this is important

This ensures that:

- both models see the same training proteins
- both models are evaluated on the same validation proteins
- any observed performance difference can be attributed more confidently to the feature representation rather than the data split

In [3]:
```
train_idx = np.load("data_processed_kmer/train_idx.npy")
val_idx   = np.load("data_processed_kmer/val_idx.npy")

X_train = X_esm[train_idx]
X_val   = X_esm[val_idx]

if hasattr(Y, "tocsr"):   # sparse
    Y_train = Y[train_idx]
    Y_val   = Y[val_idx]


print("X_train:", X_train.shape)
print("X_val:  ", X_val.shape)
print("Y_train:", Y_train.shape)
print("Y_val:  ", Y_val.shape)
```

```
X_train: (112455, 640)
X_val:   (28114, 640)
Y_train: (112455, 31454)
Y_val:   (28114, 31454)
```

Step 3. Standardize the ESM Embeddings

Why this step?

Although ESM embeddings are pretrained representations, their numerical scales can still vary across dimensions.

Applying feature standardization can improve optimization stability by:

- centering each embedding dimension
- normalizing the scale across features
- making MLP training more stable and efficient

Important design choice

The scaler is fit **only on the training set** and then applied to the validation set.

Why this matters

Fitting the scaler on all data would leak validation information into training.
Using training-only statistics preserves a proper supervised learning pipeline.

In [4]:
```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train).astype(np.float32)
X_val   = scaler.transform(X_val).astype(np.float32)
```

In [5]:
```
with open("esm_scaler.pkl", "wb") as f:
    pickle.dump(scaler, f)
```

Step 4. Build a Sparse-Safe PyTorch Dataset

Why this step?

The GO label matrix is extremely large and sparse, especially in the full-label setting.

If we convert the entire target matrix into a dense array upfront, memory usage becomes unnecessarily large because:

- the number of proteins is large
- the number of GO terms exceeds 31,000
- most labels are zeros

What this dataset does

Our custom dataset:

- returns one dense ESM embedding vector per protein
- converts only one target row at a time from sparse to dense form
- produces `(x, y)` pairs ready for mini-batch training

Why this approach is useful

This lets us train on the **full GO space** while keeping memory usage manageable.

In [6]:
```
import torch
from torch.utils.data import Dataset, DataLoader

class ProteinESMDataset(Dataset):
    def __init__(self, X, Y):
        self.X = X.astype(np.float32)
        self.Y = Y
        self.is_sparse = hasattr(Y, "tocsr")
        if self.is_sparse:
            self.Y = Y.tocsr()
            self.n_labels = self.Y.shape[1]

    def __len__(self):
        return self.X.shape[0]
```

```
    def __getitem__(self, idx):
        x = torch.from_numpy(self.X[idx])

        if self.is_sparse:
            start = self.Y.indptr[idx]
            end = self.Y.indptr[idx + 1]
            pos_idx = self.Y.indices[start:end]

            y = torch.zeros(self.n_labels, dtype=torch.float32)
            y[pos_idx] = 1.0
        else:
            y = torch.tensor(self.Y[idx], dtype=torch.float32)

        return x, y
```

Step 5. Create DataLoaders for Mini-Batch Training

Why this step?

DataLoaders allow us to iterate over proteins efficiently during training and validation.

They help us:

- shuffle the training set each epoch
- batch proteins into manageable mini-batches
- keep validation deterministic
- transfer data efficiently to GPU if available

Why batch size is kept moderate

Even though the input embedding dimension is only 640, the output dimension remains very large because the model predicts all GO terms simultaneously.
A moderate batch size helps keep training stable in memory and runtime.

In [7]:
```
train_dataset = ProteinESMDataset(X_train, Y_train)
val_dataset   = ProteinESMDataset(X_val, Y_val)

train_loader = DataLoader(train_dataset, batch_size=64, shuffle=True)
val_loader   = DataLoader(val_dataset, batch_size=64, shuffle=False)
```

Step 6. Define the ESM MLP Model

Why this step?

The model maps each protein's ESM embedding vector to a multi-label GO prediction vector.

Since ESM already provides a compact, learned protein representation, we use a multilayer perceptron (MLP) as the downstream classifier:

- simple and efficient
- appropriate for fixed-length embedding inputs
- easy to compare directly against the k-mer baseline

Model structure

The architecture uses:

- two hidden layers
- batch normalization for stability
- ReLU nonlinearities
- dropout for regularization

Why this is a meaningful baseline

Compared with k-mer features, ESM embeddings carry richer sequence information learned from large-scale pretraining.
This makes the ESM-MLP model a strong sequence-only baseline before introducing structure-based models.

In [8]:
```
import torch.nn as nn

class ESMMLP(nn.Module):
    def __init__(self, input_dim, output_dim, hidden1=512, hidden2=256, drop
        super().__init__()
        self.net = nn.Sequential(
            nn.Linear(input_dim, hidden1),
            nn.BatchNorm1d(hidden1),
            nn.ReLU(),
            nn.Dropout(dropout),

            nn.Linear(hidden1, hidden2),
            nn.BatchNorm1d(hidden2),
            nn.ReLU(),
            nn.Dropout(dropout),

            nn.Linear(hidden2, output_dim)
        )

    def forward(self, x):
        return self.net(x)
```

Step 7. Initialize the Model on the Available Device

Why this step?

Here we:

- detect the available computation device
- construct the model using the correct input and output dimensions
- move the model to CPU or GPU
- inspect the total parameter count

Why parameter count matters

Although the input dimension is smaller than k-mer, the final output layer is still very large because the model predicts all GO terms at once.
Printing the number of parameters helps us understand the computational size of the full-label classifier.

In [9]:
```
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")

model = ESMMLP(
    input_dim=X_train.shape[1],
    output_dim=Y_train.shape[1]
).to(device)

print(model)
```

```
ESMMLP(
  (net): Sequential(
    (0): Linear(in_features=640, out_features=512, bias=True)
    (1): BatchNorm1d(512, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
    (2): ReLU()
    (3): Dropout(p=0.3, inplace=False)
    (4): Linear(in_features=512, out_features=256, bias=True)
    (5): BatchNorm1d(256, eps=1e-05, momentum=0.1, affine=True, track_runnin
g_stats=True)
    (6): ReLU()
    (7): Dropout(p=0.3, inplace=False)
    (8): Linear(in_features=256, out_features=31454, bias=True)
  )
)
```

```
/global/home/users/yushanfu/esm_env/lib/python3.11/site-packages/torch/cuda/
__init__.py:180: UserWarning: CUDA initialization: The NVIDIA driver on your
system is too old (found version 12030). Please update your GPU driver by do
wnloading and installing a new version from the URL: http://www.nvidia.com/D
ownload/index.aspx Alternatively, go to: https://pytorch.org to install a Py
Torch version that has been compiled with your version of the CUDA driver.
(Triggered internally at /pytorch/c10/cuda/CUDAFunctions.cpp:119.)
  return torch._C._cuda_getDeviceCount() > 0
```

Step 8. Compute Positive Class Weights for Imbalanced Multi-Label Learning

Why this step?

GO prediction is an extremely imbalanced multi-label problem.
Most entries in the target matrix are zeros, and many GO terms appear only rarely across proteins.

The preprocessing report shows that the GO label matrix is highly sparse and follows a strong long-tail frequency distribution. :contentReference[oaicite:2]{index=2}

What we do

We compute a `pos_weight` value for each GO term so that:

- rare labels receive larger weights
- common labels receive smaller weights
- labels with zero or near-zero frequency do not create numerical instability

Why clipping is necessary

Without clipping, very rare labels can produce extremely large weights, which may cause the training or validation loss to explode.
Clipping keeps optimization numerically stable while still addressing label imbalance.

In [10]:
```
import torch.optim as optim

if hasattr(Y_train, "sum") and not isinstance(Y_train, np.ndarray):
    label_counts = np.asarray(Y_train.sum(axis=0)).ravel().astype(np.float32
else:
    label_counts = Y_train.sum(axis=0).astype(np.float32)

n_train = Y_train.shape[0]

# Compute positive class weights for each GO term
pos_weight = (n_train - label_counts) / (label_counts + 1e-6)

# Clip extreme values for stability
pos_weight = np.clip(pos_weight, 1.0, 100.0)

print("label_counts shape:", label_counts.shape)
print("pos_weight min/max:", pos_weight.min(), pos_weight.max())

pos_weight_tensor = torch.tensor(pos_weight, dtype=torch.float32, device=dev

criterion = nn.BCEWithLogitsLoss(pos_weight=pos_weight_tensor)
optimizer = optim.Adam(model.parameters(), lr=1e-3, weight_decay=1e-5)
```

```
label_counts shape: (31454,)
pos_weight min/max: 1.0 100.0
```

Step 9. Define Training and Validation Loops

Why this step?

To train the model reproducibly and clearly, we separate the workflow into:

- a training loop with gradient updates
- a validation loop without gradient updates

Why this separation helps

This makes the notebook easier to:

- debug
- reuse across experiments
- compare training and validation behavior
- save the best model checkpoint based on validation performance

What happens during training

For each mini-batch:

1. move embeddings and targets to the device
2. compute logits
3. compute weighted multi-label loss
4. backpropagate gradients
5. update model parameters

In [11]:
```
def train_one_epoch(model, loader, criterion, optimizer, device):
    model.train()
    total_loss = 0.0

    for X_batch, Y_batch in loader:
        X_batch = X_batch.to(device)
        Y_batch = Y_batch.to(device)

        optimizer.zero_grad()
        logits = model(X_batch)
        loss = criterion(logits, Y_batch)
        loss.backward()
        optimizer.step()

        total_loss += loss.item()

    return total_loss / len(loader)


@torch.no_grad()
def evaluate_loss(model, loader, criterion, device):
    model.eval()
    total_loss = 0.0
```

```
    for X_batch, Y_batch in loader:
        X_batch = X_batch.to(device)
        Y_batch = Y_batch.to(device)

        logits = model(X_batch)
        loss = criterion(logits, Y_batch)

        total_loss += loss.item()

    return total_loss / len(loader)
```

Step 10. Collect Validation Predictions

Why this step?

Loss values alone do not fully describe model quality in a multi-label setting.
To evaluate prediction quality, we need probability outputs for every GO term on the validation set.

What this step returns

For all validation proteins, we collect:

- predicted probabilities after sigmoid
- true GO label vectors

Why probabilities are needed

A fixed threshold such as 0.5 is often suboptimal in highly imbalanced multi-label tasks.
Storing predicted probabilities allows us to sweep across thresholds and evaluate different operating points.

In [12]:
```
@torch.no_grad()
def get_predictions(model, loader, device):
    model.eval()
    all_probs = []
    all_targets = []

    for X_batch, Y_batch in loader:
        X_batch = X_batch.to(device, non_blocking=True)

        logits = model(X_batch)
        probs = torch.sigmoid(logits).cpu().numpy()

        all_probs.append(probs)
        all_targets.append(Y_batch.numpy())

    all_probs = np.vstack(all_probs)
    all_targets = np.vstack(all_targets)
```

```
    return all_probs, all_targets
```

Step 11. Evaluate Performance Across Multiple Thresholds

Why this step?

Each GO term is predicted as a probability, so we must choose a threshold to convert probabilities into binary predictions.

Because the output space is sparse and highly imbalanced, the best threshold is not guaranteed to be 0.5.

Why threshold sweeping matters

The CAFA evaluation philosophy emphasizes threshold-based F-score optimization. Your preprocessing report also notes that the official metric is based on threshold sweeping, such as weighted F-max. :contentReference[oaicite:3]{index=3}

Metrics reported

We compute:

- **micro-F1**: emphasizes overall prediction performance
- **macro-F1**: emphasizes performance across all labels equally, including rare labels

In [13]:
```
from sklearn.metrics import f1_score
import numpy as np

def threshold_sweep(y_true, y_prob, thresholds=None):
    if thresholds is None:
        thresholds = np.arange(0.05, 0.55, 0.05)

    records = []

    for t in thresholds:
        y_pred = (y_prob >= t).astype(np.int8)

        micro = f1_score(y_true, y_pred, average="micro", zero_division=0)
        macro = f1_score(y_true, y_pred, average="macro", zero_division=0)

        records.append({
            "threshold": float(t),
            "micro_f1": float(micro),
            "macro_f1": float(macro)
        })

        print(f"t={t:.2f} | micro-F1={micro:.4f} | macro-F1={macro:.4f}")
```

```
    best_micro_row = max(records, key=lambda x: x["micro_f1"])
    best_macro_row = max(records, key=lambda x: x["macro_f1"])

    return records, best_micro_row, best_macro_row
```

Step 12. Train the ESM MLP Model

Why this step?

This is the main optimization stage of the notebook.

For each epoch, we:

1. train on the ESM embedding training set
2. compute validation loss
3. record both losses for later visualization
4. save the best checkpoint based on validation loss

Why save the best checkpoint

The best validation-loss model is usually more reliable than simply using the last epoch model, especially if overfitting begins during training.

Why start with a small number of epochs

The full GO output space is computationally expensive.
It is practical to first confirm that:

- the pipeline runs correctly
- the loss behaves reasonably
- checkpoints are saved properly before extending to longer runs.

In [14]:
```
num_epochs = 3
save_path = "best_esm_mlp_full_go.pt"

train_losses = []
val_losses = []

best_val_loss = float("inf")

for epoch in range(num_epochs):
    train_loss = train_one_epoch(model, train_loader, criterion, optimizer,
    val_loss = evaluate_loss(model, val_loader, criterion, device)

    train_losses.append(train_loss)
    val_losses.append(val_loss)

    print(f"Epoch {epoch+1:02d}/{num_epochs} | train_loss={train_loss:.6f} |
```

```
    if val_loss < best_val_loss:
        best_val_loss = val_loss
        torch.save(model.state_dict(), save_path)
        print(f"  -> saved best model to {save_path}")
```

```
Epoch 01/3 | train_loss=0.117373 | val_loss=0.100426
  -> saved best model to best_esm_mlp_full_go.pt
Epoch 02/3 | train_loss=0.101162 | val_loss=0.096069
  -> saved best model to best_esm_mlp_full_go.pt
Epoch 03/3 | train_loss=0.098751 | val_loss=0.094963
  -> saved best model to best_esm_mlp_full_go.pt
```

Step 13. Visualize the Training Curve

A loss curve provides a quick summary of optimization behavior.

By plotting training and validation loss across epochs, we can inspect:

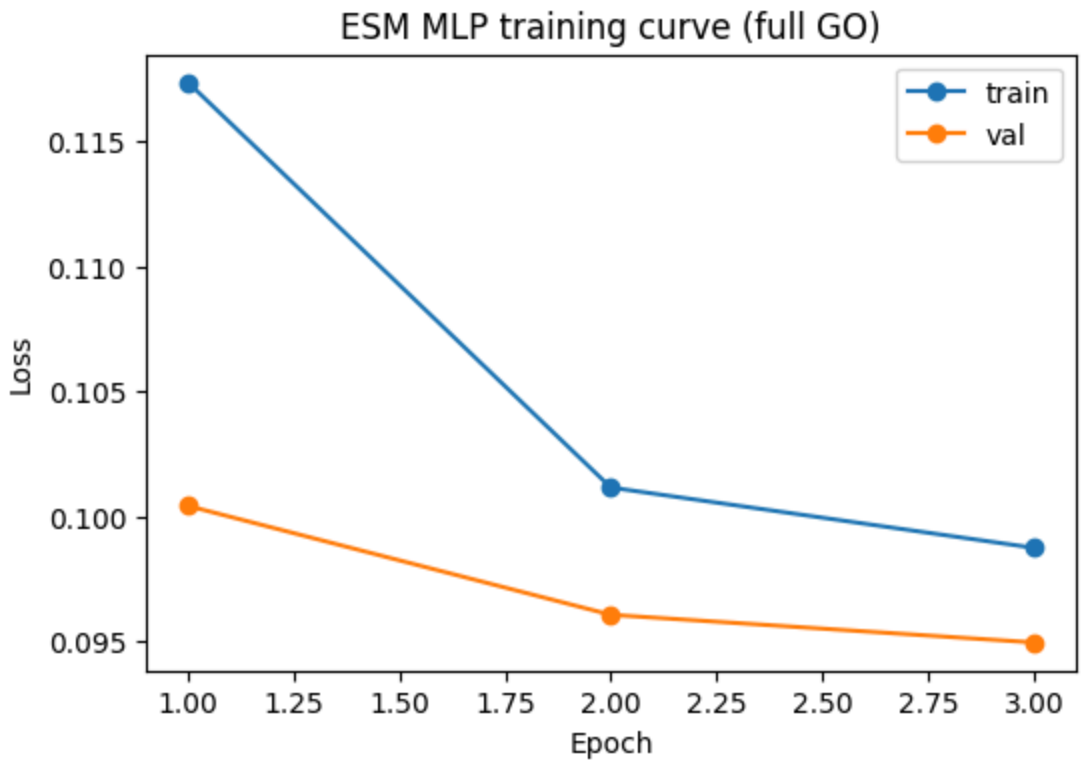
- whether the model is learning
- whether validation performance improves
- whether overfitting may be starting
- whether the optimization is stable

What to look for

- training loss should generally decrease
- validation loss should ideally decrease or stabilize
- a widening gap between training and validation loss may suggest overfitting

In [15]:
```
import matplotlib.pyplot as plt

plt.figure(figsize=(6, 4))
plt.plot(range(1, num_epochs + 1), train_losses, marker="o", label="train")
plt.plot(range(1, num_epochs + 1), val_losses, marker="o", label="val")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.title("ESM MLP training curve (full GO)")
plt.legend()
plt.show()
```



Step 14. Reload the Best Checkpoint

Why this step?

Instead of evaluating the final epoch directly, we reload the checkpoint with the best validation loss.

Why this matters

This ensures that all downstream evaluation steps are based on the strongest saved version of the model, which is especially important for fair comparison against the k-mer baseline and future models.

In [16]:
```
best_model = ESMMLP(
    input_dim=X_train.shape[1],
    output_dim=Y_train.shape[1]
).to(device)

best_model.load_state_dict(torch.load(save_path, map_location=device))
```

Out[16]:
```
<All keys matched successfully>
```

Step 15. Generate Validation Predictions with the Best Model

Why this step?

After reloading the best checkpoint, we run the model on the validation set to obtain:

- predicted GO probabilities
- true GO labels

Why this is necessary

These predictions are required for threshold sweeping and final multi-label F1 evaluation.

In [17]:
```
val_probs, val_targets = get_predictions(best_model, val_loader, device)

print("val_probs shape:", val_probs.shape)
print("val_targets shape:", val_targets.shape)
```

```
val_probs shape: (28114, 31454)
val_targets shape: (28114, 31454)
```

Step 16. Run Threshold Sweep and Identify Best Validation Thresholds

Why this step?

This step evaluates how prediction quality changes across multiple probability thresholds.

Rather than assuming one threshold in advance, we search across several candidate values and identify:

- the threshold with the best micro-F1
- the threshold with the best macro-F1

Why both thresholds are informative

- **Best micro-F1** reflects large-scale overall prediction quality
- **Best macro-F1** reflects behavior on the long-tail GO labels

In a highly imbalanced task, these may differ substantially.

In [18]:
```
thresholds = np.arange(0.05, 0.55, 0.05)
records, best_micro_row, best_macro_row = threshold_sweep(val_targets, val_p

print("\nBest by micro-F1:", best_micro_row)
print("Best by macro-F1:", best_macro_row)
```

```
t=0.05 | micro-F1=0.0217 | macro-F1=0.0065
t=0.10 | micro-F1=0.0331 | macro-F1=0.0087
t=0.15 | micro-F1=0.0429 | macro-F1=0.0100
t=0.20 | micro-F1=0.0521 | macro-F1=0.0113
t=0.25 | micro-F1=0.0612 | macro-F1=0.0125
t=0.30 | micro-F1=0.0703 | macro-F1=0.0135
t=0.35 | micro-F1=0.0797 | macro-F1=0.0145
t=0.40 | micro-F1=0.0893 | macro-F1=0.0154
t=0.45 | micro-F1=0.0993 | macro-F1=0.0161
t=0.50 | micro-F1=0.1091 | macro-F1=0.0162

Best by micro-F1: {'threshold': 0.5, 'micro_f1': 0.10912382285353502, 'macro
_f1': 0.016195805188390765}
Best by macro-F1: {'threshold': 0.5, 'micro_f1': 0.10912382285353502, 'macro
_f1': 0.016195805188390765}
```

Step 17. Save the Experiment Results

Saving the main outputs makes the experiment reproducible and easier to compare against:

- the k-mer baseline
- future ESM variants
- namespace-specific models
- structure-based approaches

What we save

We store:

- model name
- input and output dimensions
- batch size and epoch count
- best validation loss
- best threshold-based F1 results
- training and validation loss history

Why this is useful

This creates a compact experiment record that can be reused in reports, comparison tables, and future notebooks.

In [21]:
```
import json
batch_size = 64
results = {
    "model": "esm_mlp_full_go",
    "input_dim": int(X_train.shape[1]),
    "output_dim": int(Y_train.shape[1]),
    "batch_size": batch_size,
```

```
    "num_epochs": num_epochs,
    "best_val_loss": float(best_val_loss),
    "best_micro": best_micro_row,
    "best_macro": best_macro_row,
    "train_losses": train_losses,
    "val_losses": val_losses,
}

with open("esm_mlp_full_go_metrics.json", "w") as f:
    json.dump(results, f, indent=2)

print("Saved metrics to esm_mlp_full_go_metrics.json")
```

```
Saved metrics to esm_mlp_full_go_metrics.json
```

Step 18. Summarize Threshold Sweep Results in Table Form

Why this step?

A table makes it easy to inspect how prediction quality changes with the decision threshold.

What this helps us understand

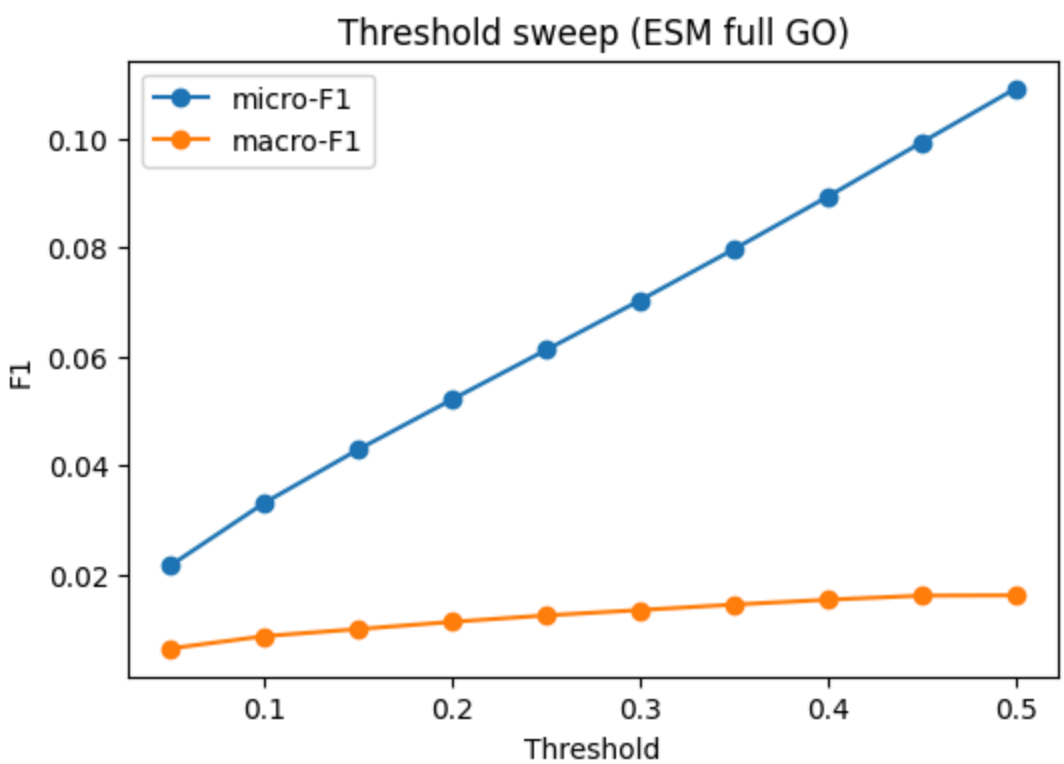
- whether performance is stable across nearby thresholds
- where the best operating point occurs
- whether micro-F1 and macro-F1 peak at similar or different thresholds

In [22]:
```
import pandas as pd

df_thresh = pd.DataFrame(records)
display(df_thresh)
```

	threshold	micro_f1	macro_f1
0	0.05	0.021654	0.006459
1	0.10	0.033132	0.008710
2	0.15	0.042923	0.010012
3	0.20	0.052108	0.011334
4	0.25	0.061173	0.012516
5	0.30	0.070282	0.013499
6	0.35	0.079655	0.014503
7	0.40	0.089312	0.015392
8	0.45	0.099252	0.016146
9	0.50	0.109124	0.016196

In [23]:
```
plt.figure(figsize=(6, 4))
plt.plot(df_thresh["threshold"], df_thresh["micro_f1"], marker="o", label="m
plt.plot(df_thresh["threshold"], df_thresh["macro_f1"], marker="o", label="m
plt.xlabel("Threshold")
plt.ylabel("F1")
plt.title("Threshold sweep (ESM full GO)")
plt.legend()
plt.show()
```



In [ ]: